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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-DS-006**

1. Topic

AI-Based Stress-Aware VR Meditation System for Pain Reduction

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	**R26-DS-006**
Year	**2026**

Research Problem

Contemporary society is typified by the high proportion of people that show common mental and physical illnesses like stress, headache, mental exhaustion, depression and sudden anger. The general population, working professionals, and students are often susceptible to these conditions, and they are unable to focus, manage their emotions, and lead a good life. Such disorders influence psychological health, interpersonal relationships and productivity to a great extent though they do not always involve medical treatment. The main consumers of the proposed solution are healthy people who have discomfort due to stress and cannot concentrate on their daily routine.

The discomfort is closely associated with the attention as the information on psychological studies and personal meditation experience demonstrates. The pain or stress is subjectively perceived when there is a lot of attention to the pain. On the other hand, the perceived level of discomfort tends to decrease because of distracting the attention with conscious breathing, relaxing music or seeing an image or video. This effect has been demonstrated by mindfulness and meditation practices that decrease the perception of stress and discomfort through training people to be aware of sensations without being obsessed. However, the situational awareness, discipline or training to implement these strategies in a pressure situation is lacking in most people.

It is not just an issue of individual wellbeing. Academic achievement is reduced; work is inefficient, emotional instability and high long-term societal and healthcare spending are all linked to chronic stress. Technically, there is an apparent lack of systems that can detect physiological manifestations of stress and react to them in real time in an adaptive manner. Morally and legally, there is a desperate need for non-invasive, non-clinical solutions which would improve mental well-being without resorting to hospital-based interventions and a medical claim.

The paper is an intersection between data science, affective computing, and immersive technologies and their use in wellness, mental health support, and human-centered artificial intelligence. The project is in line with the Sustainable Development Goals of the United Nations, especially SDG 3: Good Health and Well-Being through the promotion of preventive, accessible, and technology-assisted approaches to mental wellness. The proposed study is expected to offer a scalable, ethical, and socially responsible solution to the issue by conducting research on routine stress and attention control within a non-clinical setting.

Existing Research and Systems

Numerous existing studies and systems have been developed to help with pain and stress relief. Most of these approaches utilize either Virtual Reality (VR) technology, meditation-based therapies or physiological biofeedback as independent tools, or a combination of the above. There are also many commercial and research-oriented VR applications that provide users with immersive nature scenes or guided meditation experiences for relaxation and pain distraction. However, most of these systems are not dynamic and instead, the user is given a pre-programmed environment without any type of real-time physiological adaptation to the user's changing state.

Numerous studies have shown that VR-assisted meditation significantly reduces reported stress and pain levels, especially in patients with chronic pain or stress-related conditions [1]. Similarly, the use of VR-based HRV biofeedback systems has shown to be effective for relaxation and emotional regulation by assisting users through paced breathing exercises while in an immersive environment [2]. However, while in this approach, users have access to their physiological information, such as heart rate, this information is generally only being collected and used as either an outcome measure of success, or explicit feedback to the user.

Recent studies on the use of adaptive virtual reality systems using both physiological signals and machine learning focused primarily on training and cognitive load management [3]. This technology provides a dynamic adjustment to environmental complexity based on stress and cognitive workload of the user. Although these advanced technologies are not presently applied to reducing the pain associated with using meditation and typically require multiple sensors that limit accessibility, they are also not scalable.

The limitations of existing research and products include three primary areas. [1] There is no existing closed-loop VR system that utilizes machine learning to allow for automatic adjustments to the virtual environment based on stress inference. [2] Minimal investigation has been conducted into creating systems that consider individual physiological baselines and user preferences for personalization. [3] Most of the available systems are based on medical-grade hardware and therefore are not capable of being used in everyday applications. These three gaps present the opportunity for a personalized, adaptive, data-driven VR system that can incorporate the use of wearable technology to provide widespread accessibility to distraction from pain through meditation.

The Solution and The Conceptual Diagram

The proposed solution consists of a closed-loop, data-driven VR meditation system that keeps track of the user's heart-rate signals and adjusts the virtual stimuli to reduce the user's stress and pain feel through the sensed skin. The system is based on an Input-Process-Output model: the input consists of the data coming from a consumer wearable device in the form heart rate; in the process stage the data gets signal conditioning, feature extraction and ML inference

which finally gets a momentary stress metric; the output is a VR meditation environment that is adaptively parametrized to the user in terms of the sensory elements (visual, auditory, haptic and guided breathing cues) which are regulated in real time to make the user experience both physiological and subjective relaxation.

Signal acquisition and preprocessing are used to acquire the time domain and frequency domain (e.g., RMSSD, SDNN, LF/HF) heart-rate variability (HRV) features along with simple statistical descriptors. These features after processing it, get rid of the noise in the signal and allow the use of the high-latency inputs for the ML-based stress inference module while still being able to run on consumer hardware. The system thus should get preference for scalability and ecological validity.

The stress inference based on ML method processes the HR/HRV feature data in short time windows, to provide as the output a discrete stress class (low/medium/high) or a continuous stress score which is suitable for closed-loop control. The possible algorithms that can be used are tree-ensemble classifiers from classical methods and temporal models (for instance, LSTM) for sequential smoothing; the research activity has already included a systematic comparison of generalized versus personalized models and an exploration of lightweight on-device inference versus server-assisted inference to pretty much equally distribute latency, accuracy, and privacy.

The adaptive VR meditation module connects guess stress levels with a limited range of environment parameters: the complexity of the scene, temperature, movement amplitude, audio and vocal guidance speed. The rules for adaptation are at first cautious and understandable (for example, to cut movement and slow down audio when the user is under stress) and then further refined with the help of the personalization layer. The process of adjusting is done in an iterative way and has the goal of user comfort (minimizing feelings of nausea due to the simulator) being the main one while, at the same time, relaxation effect being maximized.

The loop is completed by a personalization and evaluation layer. Personalization uses the first desktop questionnaire (preferences, prior meditation experience) and a short baseline calibration to define individualized thresholds and adaptation sensitivity. Continuous data from physiological sensors and subjective reports of outcomes provide the basis for constant modeling.

fine-tuning and parameter adaptation. Evaluation consists of within-subject statistical testing (pre/post measures, paired tests) and effect-size reporting to confirm the system's effect on stress and pain perception.

This architecture directly tackles the shortcomings mentioned in the literature: it uses real-time ML-based adaptation instead of merely choosing scenes statically, offers personalization for individuals regarding both model and VR parameters, and keeps the consumer wearables in the

loop to ensure maximum accessibility and deployment potential [1]–[3]. Consequently, the system poses a practical, experimentally testable closed-loop approach to meditation induced pain and stress reduction that is fit for the research while providing clear routes to further clinical evaluation.

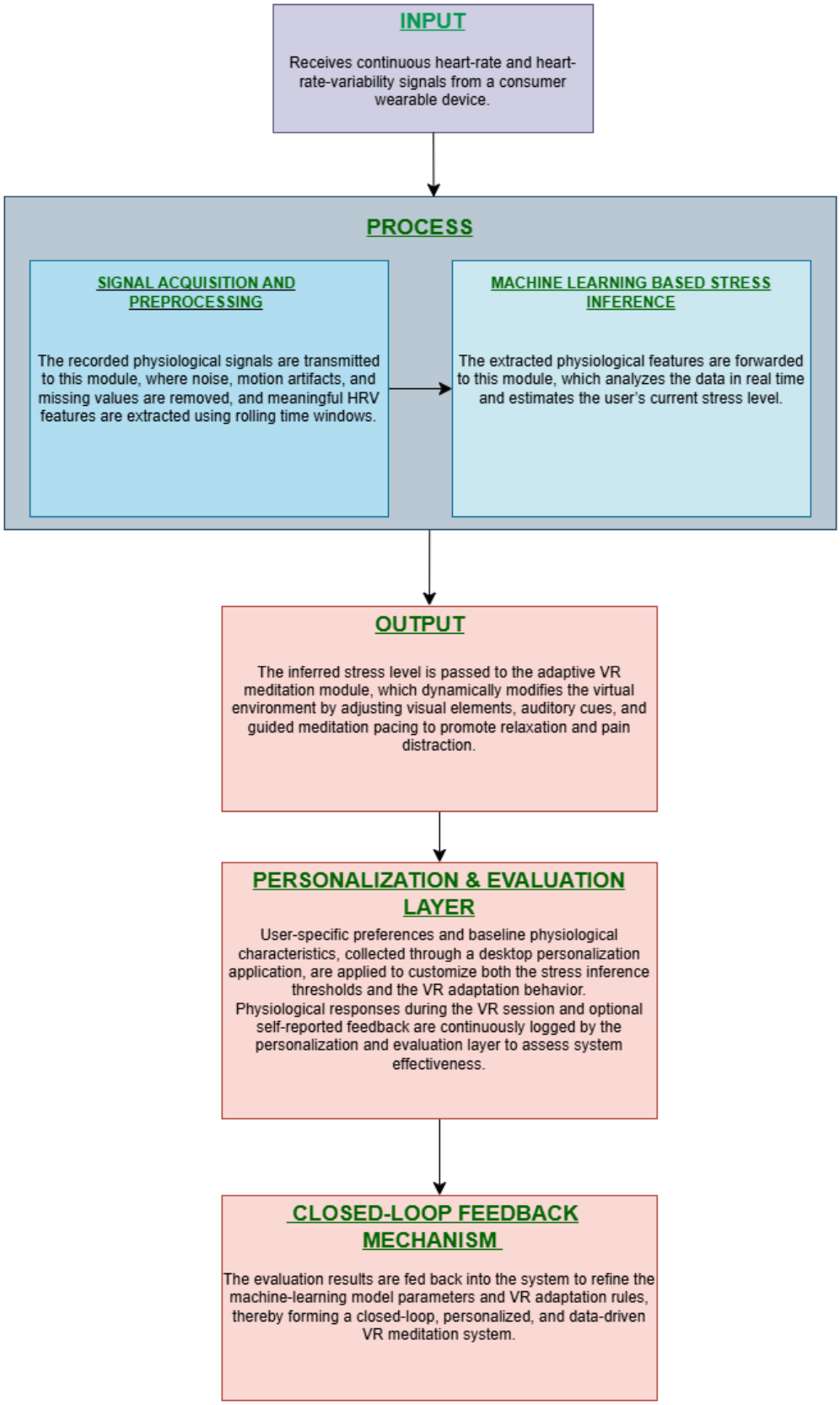


Figure 1: Conceptual Diagram

Availability of Necessary Specialized Domain Expertise, Knowledge, and Data.

Gaining Specialized Domain Expertise

Our project requires understanding across four domains: meditation science, stress physiology, adaptive systems design, and VR wellness applications.

Meditation and Pain Psychology: Team members will consult with meditation instructors from local wellness centers in Colombo and review established research on mindfulness-based pain reduction. We'll study how attention redirection works, why meditation reduces perceived pain, and what environmental factors support deep meditative states. Resources include published research on meditation neuroscience and pain psychology available through Google Scholar and PubMed.

Heart Rate and Stress Correlation: We'll gain expertise in understanding how heart rate patterns reflect stress levels by studying biofeedback literature and sports science research. This includes learning about Heart Rate Variability (HRV) metrics, stress response patterns, and how to distinguish meditation-induced relaxation from general calmness. Online courses on bio signal processing and wellness technology will provide this foundation.

Personalization Systems: The team member handling Component 4 will study recommender systems, user modeling techniques, and adaptive learning algorithms. This involves understanding how Netflix-style recommendation engines work and how to apply similar principles to meditation experiences. Research papers on personalized health interventions and adaptive user interfaces will guide this learning.

VR Wellness Design: We'll explore existing VR meditation applications and study design principles for calming immersive environments. This includes understanding what makes VR experiences comfortable versus nauseating, optimal audio-visual combinations for relaxation, and how to implement smooth environmental transitions that don't break immersion.

Gaining Technical Knowledge and Skills

Each component requires specific technical skills that team members will develop through structured learning:

Component 1 (Biofeedback Acquisition): Learning smartwatch SDK integration, Bluetooth data streaming, real-time signal processing using Python libraries like HeartPy and NeuroKit2, and techniques for handling noisy bio signal data. Online tutorials and device documentation will be primary resources.

Component 2 (Stress Prediction): Developing machine learning skills for time-series classification, feature engineering from HRV data, and model training using scikit-learn or TensorFlow. Kaggle

competitions on bio signal analysis and online courses on applied ML for healthcare data will build these capabilities.

Component 3 (Adaptive VR): Learning Unity3D or Unreal Engine through official tutorials, implementing real-time parameter adjustment systems, designing calming 3D environments, and integrating spatial audio. Unity Learn platform and YouTube channels focused on VR development provide comprehensive guidance.

Component 4 (Personalization & Analytics): Building recommender systems, implementing simple reinforcement learning for parameter optimization, creating data visualization dashboards using libraries like Plotly or D3.js, and developing user behavior analysis pipelines. Online courses on recommendation systems and adaptive algorithms are readily available on Coursera and Udacity.

All team members will engage in weekly knowledge-sharing sessions to understand how components integrate, ensuring everyone grasp the full system architecture.

Data Collection Strategy

Our data collection follows a multi-phase approach designed to work within Sri Lankan privacy regulations:

Phase 1 - Public Datasets: We'll begin with freely available datasets including stress and relaxation bio signal databases from PhysioNet, meditation EEG datasets, and published HRV data during various mental states. These provide training data for initial stress prediction models without requiring sensitive data collection.

Phase 2 - Volunteer Testing Sessions: We'll recruit 20-30 university students and staff as volunteers for controlled testing sessions. Each session involves:

• Wearing a consumer heart rate monitor for 10-15 minute VR meditation sessions

• Self-reporting pain/discomfort levels before and after (using 0-10 scales)

• Rating relaxation quality and environment preferences

• Providing feedback on which VR settings felt most calm

Volunteers will be recruited through campus announcements with clear information about the study's wellness focus. All participation is voluntary with written consent.

Phase 3 - Session Outcome Data: As Component 4's focus, we'll systematically collect structured session logs including initial stress levels, VR parameters used, duration, final stress levels, user satisfaction ratings, and pain perception changes. This creates a rich dataset for training personalization algorithms and analyzing what environmental adaptations work best.

Phase 4 - Synthetic Data Augmentation: To expand our dataset, we'll generate synthetic user profiles and session outcomes using data augmentation techniques, allowing us to test edge cases and rare user response patterns.

All collected data will be anonymized immediately, with no names or identifying information stored. Data will be encrypted and kept on university-approved secure servers with access limited to team members.

Ethical Clearance Requirements

Yes, this project requires ethical approval, though with reduced complexity compared to clinical medical research.

We'll submit our proposal to the university's Ethics Review Committee, positioning this as wellness technology research rather than medical treatment. Our submission will cover:

Informed Consent: All volunteers will receive clear information sheets explaining this is a meditation technology research project, what data we collect, how it's used, their right to withdraw anytime, and that this is not medical treatment.

Risk Assessment: The primary risks are VR-induced discomfort or motion sickness. We'll mitigate this by screening participants for VR sensitivity, keeping sessions short (under 20 minutes), providing immediate stop options, and having protocols for anyone feeling unwell.

Data Privacy Compliance: Under Sri Lanka's Personal Data Protection Act No. 9 of 2022, we'll ensure minimal data collection, strong encryption, clear retention policies, and complete transparency about data usage. Participants can request data deletion at any time.

Scope Limitation: We'll explicitly state that our system is for relaxation and wellness support, not for treating diagnosed medical conditions. Any participant with severe chronic pain or mental health conditions will be advised to consult healthcare professionals rather than relying solely on our application.

The ethics review process typically takes 2-4 weeks, so we'll submit early while developing technical components, allowing parallel progress.

Objectives and Novelty

Main Objective

To design and evaluate an adaptive virtual reality meditation system that utilizes machine learning–based stress prediction from heart rate data to dynamically modify the VR environment, with the aim of supporting perceived pain reduction through personalized and data-driven interventions.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Perera W.T.S.R IT22231314	Develop an adaptive VR meditation environment that responds dynamically to predicted stress levels	• Design adaptive VR environment with adjustable parameters • Map predicted stress levels to environmental adaptations • Implement real-time VR adaptation logic	The VR environment dynamically adapts based on inferred physiological stress rather than remaining static, enabling a closed-loop adaptive experience that is underexplored in meditation-based VR pain distraction research.
Perera B.D.P.D IT22191038	Design and implement machine learning models to predict user stress levels from physiological data	• Define stress labels and prediction targets • Train and evaluate multiple machine learning models • Perform feature selection and hyperparameter tuning	Introducing machine learning–based stress inference as a control signal for adaptive VR meditation, moving beyond static threshold-based biofeedback approaches commonly found in existing systems

Samarasekara K.S.M IT22172396	Develop a physiological data acquisition and preprocessing module for stress-related feature extraction	• Collect heart rate data using a consumer wearable device • Process raw physiological signals to handle noise and artifacts • Extract relevant HR and HRV feature for stress analysis • Prepare a structured dataset for machine learning model training	Unlike many existing systems that rely on clinical-grade sensors, this component focuses on extracting meaningful stress-related features from consumer-grade wearable devices, improving accessibility and scalability
Prathikesh S. IT22621788	Integrate personalization and evaluate the system's effectiveness in stress and pain modulation	• Develop a desktop application for user preference collection and baseline calibration • Integrate all system components into a unified pipeline • Design and conduct user experiments • Perform statistical analysis on stress and perceived pain outcomes	This component introduces multi-level personalization, combining user preferences and physiological baselines and provides quantitative evaluation of adaptive versus non-adaptive VR meditation, which is rarely addressed in existing research

5. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Samadhi	Rathnayake	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

References

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[3] Nasri, M. (2025). A Physiological Adaptation Framework for Cognitive Load and Stress Detection in Virtual Reality. arXiv preprint arXiv:2504.06461.